



Home > Science 6: Section 2 > Topic 4 - Introduction to Earth's Systems > Topic 4 Test > Attempt 1

Topic 4 Test

ATTEMPT SCORE

34 / 40

01 True/False

2 / 2

True or false.

The majority of water on earth is freshwater.

☐ True

☒ False



Student's Correction

The majority of water on earth is saltwater. Only 3% of Earth's water is freshwater. And from that 3%, 70% is frozen.

Correction Guidelines

02 Matching

2 / 2

Matching.

A terrible mudslide in Regent, Sierra Leone, killed nearly 1,000 people in 2017. Part of Sugar Loaf Mountain detached and slid downslope, burying entire neighborhoods. Label each cause of the tragedy as belonging to the hydrosphere, biosphere, or geosphere.

many people had recently moved to Regent to be close to the capital, Freetown

A

biosphere



people cut the trees on the mountainside to build houses and get fuel for fires

B

biosphere



the rains in 2017 were unusually heavy

C

hydrosphere



⚙ hydrosphere

⚙ biosphere

⚙ geosphere

03 Multiple Choice

2 / 2

Choose the best answer.

Most of the freshwater is tied up in one or two different forms. In what form is 70% of Earth's fresh water found?

- ☐ freshwater lakes
- ☒ frozen fresh water
- ☐ oceans
- ☐ liquid fresh water

**04 True/False****0 / 2****True or false.**

Condensation occurs when air gains energy (increases in temperature).

- ☒ True
- ☐ False

**05 Matching****4 / 4**

Matching.

Match the image to the corresponding landform.



A

delta



B

dune





C

mountain



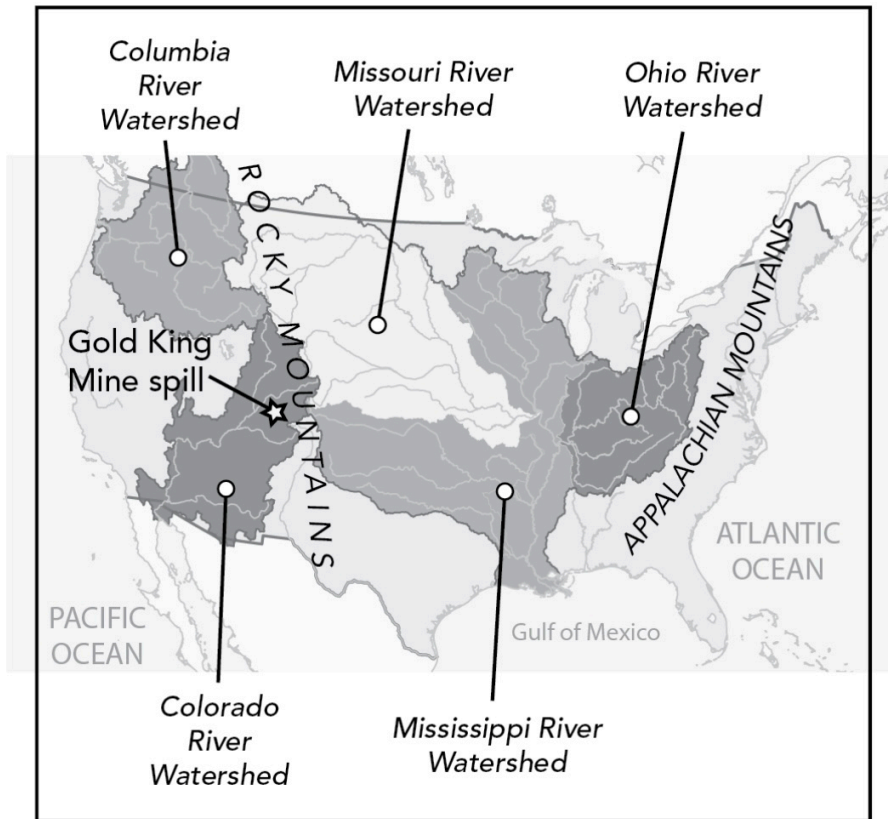
D

valley



Choose the best answer.

In 2015, an accidental release of toxic chemicals from the Gold King Mine contaminated over 3 million gallons of water. The communities in which watershed were most likely affected?



- ☐ Columbia River Watershed
- ☒ Colorado River Watershed
- ☐ Mississippi River Watershed
- ☐ Missouri River Watershed



07 Fill in the Blank Drag and Drop

1 / 2

Fill in the blanks.

Different types of landforms are found all over Earth. One landform that is found both on Earth's surface and under the sea is 1 a valley ✓ . On the other hand, 2 a glacier ✗ is only found in the ocean.

⚡ an abyssal plain

⚡ a lake

Correct answers:

2 an abyssal plain

08 Multiple Choice

2 / 2

Choose the best answer.

What is the relationship between the hydrosphere and the cryosphere?

- ☐ the hydrosphere is part of the cryosphere
- ☐ the cryosphere drives the hydrosphere
- ☒ the cryosphere is part of the hydrosphere
- ☐ the hydrosphere contains water

**09 Multiple Choice****0 / 2****Choose the best answer.**

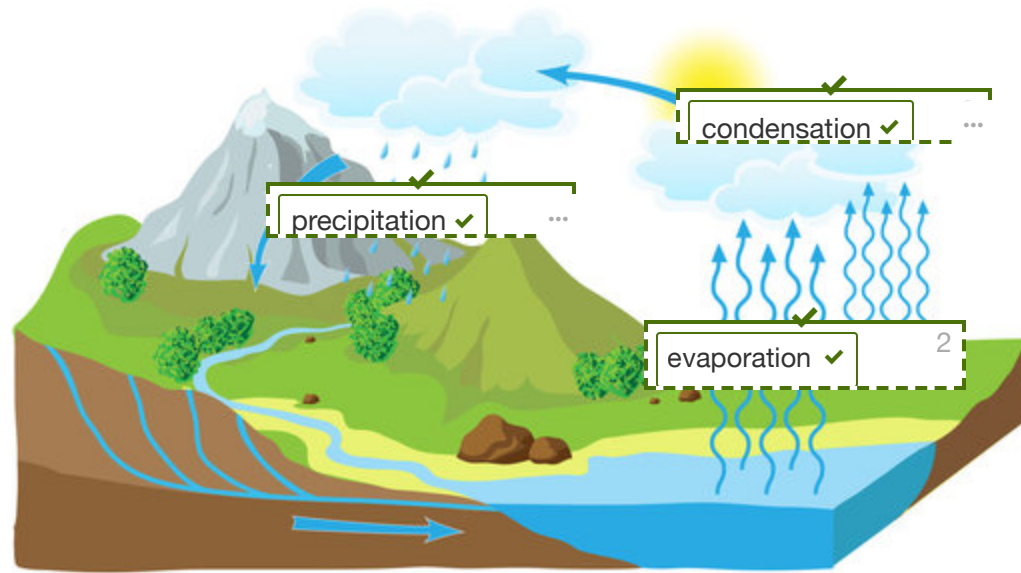
Melting glaciers are an example of positive feedback. Why is this considered positive feedback?

- ☒ Melting glaciers would mean more ice at its top, which would slow the glaciers melting.
- ☐ Melting glaciers would change the land around it, speeding up erosion.
- ☐ Melting glaciers would let the uncovered air and land warm, which speeds up the melting.
- ☐ Melting glaciers would add more water to the hydrosphere, making it wetter.

**10 Label Image****2 / 2**

Label the image.

Complete the water cycle.



Given responses:

1 precipitation ✓

2 evaporation ✓

3 condensation ✓

11 True/False

2 / 2

True or false.

Rivers form dunes by depositing sediment over big stretches of land.

☐ True☒ False**Student's Correction**

Wind forms dunes by depositing sediment over big stretches of land and rivers don't. When the river gets closer to less elevation the river slows down. Then the rivers doesn't have enough energy to carry sediment and it drops all the heavy sediment first. This process continues and then there are big piles of sediment in the middle of a river, forming deltas.

Correction Guidelines

12 Short Answer/Essay

2 / 2

You tell me.

What is transpiration and how does it contribute to the water cycle?

Transpiration is when water evaporates from the plant's leaves. Transpiration contributes to the water cycle because it evaporates from the plant's leaves and condenses into clouds. Every plant has a small part in transpiration. If we were to remove plants, overall, the rainfall would decrease drastically. Thus proving that transpiration is a small but important part of the hydrosphere.

59 / 100 Word Limit

13 True/False**2 / 2****True or false.**

The shape of a landform dictates the direction water flows.

☒ True☐ False**14 Short Answer/Essay****5 / 5**

Short answer.

Tyler, Carson, and their friend Throckmorton are planning to go on a hike in the nearby hills, but they can't agree on a route to take. They will be starting their hike from the same spot in the bottom left corner of the map. Carson's proposed route is in blue, Tyler's red, and Throckmorton's yellow.



Describe each of the proposed routes and where they will end up. In your opinion, whose route will result in the best hike? Support your answer with reasoning.

This map is a topography map which shows the elevation of all three hikers. Carson proposed a route through, probably, some plains with a gradual upwards slope. Evidences that prove this is that as you can see the lines are maintaining a gradual distance from each other, with a closer distance near the end of the hike. But a key factor to remember is that the lines have a small label labeling the elevation feet. The elevation is about 650 and Carson started at 550 so the hike had an elevation of about 100 ft. On the other hand Tyler's route is very treacherous and has a big uphill slope. I can figure this out because the lines are very close together and this proves that the slope is a great deal and is very tiring. The elevation also increases 100 ft. But, Tyler's trail also shows, based on the map, a beautiful trail next to a river. Both Carson and Tyler meet up at a plain at the summit of the mountain. Finally, Throckmorton has a trail. The trail is on the edge of the mountain and doesn't have such a great slope. Throckmorton also proposed a route on which there isn't a great elevation. As you can see, the end point of the

trail at 580 ft and the starting at 550. So the elevation of the hike is about 30 ft. This trail also has a lake on the left-side of the trail. In my opinion, Throckmorton's trail is the best one because of its gradual slope and a beautiful lake beside you.

266 / 250 Word Limit

15 Matching

5 / 5

Matching.

Match the word with the sphere that best goes with it. You may use each word more than once.

rain	A	hydrosphere	✓
mountain	B	geosphere	✓
ice	C	hydrosphere	✓
oxygen	D	atmosphere	✓
human	E	biosphere	✓
tree	F	biosphere	✓
volcano	G	geosphere	✓
grass	H	biosphere	✓
snow	I	hydrosphere	✓
carbon dioxide	J	atmosphere	✓

⚙ hydrosphere

⚙ geosphere

⚙ atmosphere

⚙ biosphere

16 Fill in the Blank Drag and Drop

1 / 2

Fill in the blanks.

Describe the following features in terms of relief and elevation:

Example - Plateau: high elevation, low relief

Mountains: 1 high ✓ elevation, 2 low ✗ relief

Plains: 3 low ✓ elevation, 4 high ✗ relief

 high low**Correct answers:**

2 high 4 low